

Random Variables

A Random Variable (RV) is a variable that probabilistically takes on a value and they are one of the most important constructs in all of probability theory. You can think of an RV as being like a variable in a programming language, and in fact random variables are just as important to probability theory as variables are to programming. Random Variables take on values, have types and have domains over which they are applicable.

Random variables work with all of the foundational theory we have built up to this point. We can define events that occur if the random variable takes on values that satisfy a numerical test (eg does the variable equal 5, is the variable less than 8).

Lets look at a first example of a random variable. Say we flip three fair coins. We can define a random variable Y to be the total number of “heads” on the three coins. We can ask about the probability of Y taking on different values using the following notation:

Let Y be the number of heads on three coin flips

$P(Y = 0) = 1/8$	(T, T, T)
$P(Y = 1) = 3/8$	(H, T, T), (T, H, T), (T, T, H)
$P(Y = 2) = 3/8$	(H, H, T), (H, T, H), (T, H, H)
$P(Y = 3) = 1/8$	(H, H, H)
$P(Y \geq 4) = 0$	

Even though we use the same notation for random variables and for events (both use capital letters) they are distinct concepts. An event is a scenario, a random variable is an object. The scenario where a random variable takes on a particular value (or range of values) is an event. When possible, I will try and use letters E,F,G for events and X,Y,Z for random variables.

Using random variables is a convenient notation technique that assists in decomposing problems. There are many different types of random variables (indicator, binary, choice, Bernoulli, etc). The two main families of random variable types are discrete and continuous. Discrete random variables can only take on integer values. Continuous random variables can take on decimal values. We are going to develop our intuitions using discrete random variable and then introduce continuous.

Properties of random variables

There are many properties of a random variable some of which we will dive into extensively. Here is a brief summary. Each random variable has:

Property	Notation Example	Description
Meaning		A semantic description of the random variable
Symbol	X	A letter used to denote the random variable
Support or Range	$\{0, 1, \dots, 3\}$	the values the random variable can take on
Distribution Function (PMF or PDF)	$P(X = x)$	A function which maps values the RV can take on to likelihood.
Expectation	$E[X]$	A weighted average

Property	Notation Example	Description
Variance	$\text{Var}(X)$	A measure of spread
Standard Deviation	$\text{Std}(X)$	The square root of variance
Mode		The most likely value of the random variable

You should set a goal of deeply understanding what each of these properties mean. There are many more properties than the ones in the table above: properties like [entropy](#), [median](#), [skew](#), [kurtosis](#).

Random variables vs Events

Random variables and events are two different concepts. An event is an outcome, or a set of outcomes, to an experiment. A random variable is a more like an experiment -- it will take on an outcome eventually. Probabilities are over events, so if you want to talk about probability in the context of a random variable, you must construct an event. You can make events by using any of the [Relational Operators](#): $<$, $\leq$, $>$, $\geq$, $=$, or $\neq$ (not equal to). This is analogous to coding where you can use relational operators to create boolean expressions from numbers.

Lets continue our example of the random variable Y which represents the number of heads on three coin flips. Here are some events using the variable Y :

Event	Meaning	Probability Statement
$Y = 1$	Y takes on the value 1 (there was one heads)	$P(Y = 1)$
$Y < 2$	Y takes on 0 or 1 (note this Y can't be negative)	$P(Y < 2)$
$X > Y$	X takes on a value greater than the value Y takes on.	$P(X > Y)$
$Y = y$	Y takes on a value represented by non-random variable y	$P(Y = y)$

You will see many examples like this last one, $P(Y = y)$, in this text book as well as in scientific and math research papers. It allows us to talk about the likelihood of Y taking on a value, in general. For example, later in this book we will derive that for three coin flips where Y is the number of heads, the probability of getting exactly y heads is:

$$P(Y = y) = \frac{0.75}{y!(3 - y)!} \quad \text{If } 0 \leq y \leq 3$$

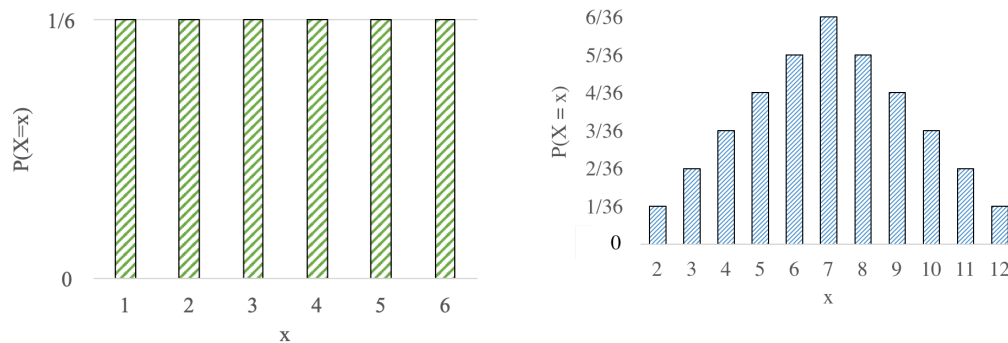
This statement above is a function which takes in a parameter y as input and returns the numeric probability $P(Y = y)$ as output. This particular expression allows us to talk about the probability that the number of heads is 0, 1, 2 or 3 all in one expression. You can plug in any one of those values for y to get the corresponding probability. It is customary to use lower-case symbols for non-random values. The use of an equals sign in the "event" can be confusing. For example what does this expression say $P(Y = 1) = 0.375$? It says that the probability that "Y takes on the value 1" is 0.375. For discrete random variables this function is called the "[probability mass function](#)" and it is the topic of our next chapter.

Probability Mass Functions

For a random variable, the most important thing to know is: how likely is each outcome? For a discrete random variable, this information is called the "**probability mass function**". The probability mass function (PMF) provides the "mass" (i.e. amount) of "probability" for each possible assignment of the random variable.

Formally, the probability mass function is a mapping between the values that the random variable could take on and the probability of the random variable taking on said value. In mathematics, we call these associations functions. There are many different ways of representing functions: you can write an equation, you can make a graph, you can even store many samples in a list. Let's start by looking at PMFs as graphs where the x -axis is the values that the random variable could take on and the y -axis is the probability of the random variable taking on said value.

In the following example, on the left we show a PMF, as a graph, for the random variable: X = the value of a six-sided die roll. On the right we show a contrasting example of a PMF for the random variable X = value of the sum of two dice rolls:



Left: the PMF of a single six-sided die roll. Right: the PMF of the sum of two dice rolls.

The [sum of two dice example](#) in the equally likely probability section. Again, the information that is provided in these graphs is the likelihood of a random variable taking on different values. In the graph on the right, the value "6" on the x -axis is associated with the probability $\frac{5}{36}$ on the y -axis. This x -axis refers to the event "the sum of two dice is 6" or $Y = 6$. The y -axis tells us that the probability of that event is $\frac{5}{36}$. In full: $P(Y = 6) = \frac{5}{36}$. The value "2" is associated with " $\frac{1}{36}$ " which tells us that, $P(Y = 2) = \frac{1}{36}$, the probability that two dice sum to 2 is $\frac{1}{36}$. There is no value associated with "1" because the sum of two dice can not be 1. If you find this notation confusing, revisit the [random variables](#) section.

Here is the exact same information in equation form:

$$P(X = x) = \frac{1}{6} \quad \text{if } 1 \leq x \leq 6 \qquad P(Y = y) = \begin{cases} \frac{(y-1)}{36} & \text{if } 1 \leq y \leq 7 \\ \frac{(13-y)}{36} & \text{if } 8 \leq y \leq 12 \end{cases}$$

As a final example, here is the PMF for Y , the sum of two dice, in Python code:

```
def pmf_sum_two_dice(y):  
    # Returns the probability that the sum of two dice is y  
    if y < 2 or y > 12:  
        return 0  
    if y <= 7:  
        return (y-1) / 36  
    else:  
        return (13-y) / 36
```

Notation

You may feel that $P(Y = y)$ is redundant notation. In probability research papers and higher-level work, mathematicians often use the shorthand $P(y)$ to mean $P(Y = y)$. This shorthand assumes that the lowercase value (e.g. y) has a capital letter counterpart (e.g. Y) that represents a random variable even though it's not written explicitly. In this book, we will often use the full form of the event $P(Y = y)$, but we will occasionally use the shorthand $P(y)$.

Probabilities Must Sum to 1

For a variable (call it X) to be a proper random variable it must be the case that if you summed up the values of $P(X = k)$ for all possible values k that X can take on, the result must be 1:

$$\sum_k P(X = k) = 1$$

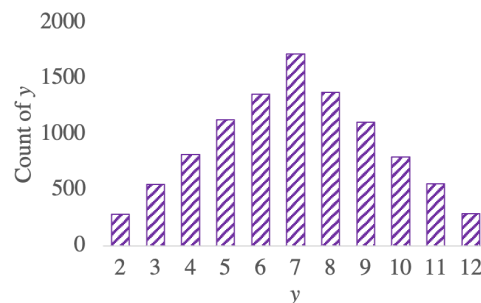
For further understanding, let's derive why this is the case. A random variable taking on a value is an event (for example $X = 2$). Each of those events is mutually exclusive because a random variable will take on exactly one value. Those mutually exclusive cases define an entire sample space. Why? Because X must take on *some* value.

Data to Histograms to Probability Mass Functions

One surprising way to store a likelihood function (recall that a PMF is the name of the likelihood function for *discrete* random variables) is simply a list of data. We simulated summing two die 10,000 times to make this example dataset:

[8, 4, 9, 7, 7, 7, 7, 5, 6, 8, 11, 5, 7, 7, 7, 6, 7, 8, 8, 9, 9, 4, 6, 7, 10, 12, 6, 7, 8, 9, 3, 7, 4, 9, 2, 8, 5, 8, 9, 6, 8, 7, 10, 7, 6, 7, 7, 5, 4, 6, 9, 5, 7, 4, 2, 11, 10, 11, 8, 4, 11, 9, 7, 10, 12, 4, 8, 5, 11, 5, 3, 9, 7, 5, 5, 5, 3, 8, 6, 11, 11, 2, 7, 7, 6, 5, 4, 6, 3, 8, 5, 8, 7, 6, 9, 4, 3, 7, 6, 6, 6, 5, 6, 10, 5, 9, 9, 8, 8, 7, 4, 8, 4, 9, 8, 5, 10, 10, 9, 7, 9, 7, 7, 10, 4, 7, 8, 4, 7, 8, 9, 11, 7, 9, 10, 10, 2, 7, 9, 4, 8, 8, 12, 9, 5, 11, 10, 7, 6, 4, 8, 9, 9, 6, 5, 6, 5, 6, 11, 7, 3, 10, 7, 3, 7, 7, 10, 3, 6, 8, 6, 8, 5, 10, 2, 7, 4, 8, 11, 8, 3, 4, 3, 8, 8, 6, 6, 12, 11, 10, 10, 10, 8, 4, 8, 4, 4, 6, 6, 7, 8]

Note that this data, on its own, represents an approximation for the probability mass function. If you wanted to approximate $P(Y = 3)$ you could simply count the number of times that "3" occurs in your data. This is an approximation based on the [definition of probability](#). Here is the full [histogram](#) of the data, a count of times each value occurs:



A normalized histogram (where each value is divided by the length of your data list) is an approximation of the PMF. For a dataset of discrete numbers, a histogram shows the count of each value (in this case y). By the definition of probability, if you divide this count by the number of experiments run, you arrive at an approximation of the probability of the event $P(Y = y)$. In our example, we have 10,000 elements in our dataset. The count of times that 3 occurs is 552. Note that:

$$\frac{\text{count}(Y = 3)}{n} = \frac{552}{10000} = 0.0552$$
$$P(Y = 3) = \frac{4}{36} = 0.0555$$

In this case, since we ran 10,000 trials, the histogram is a very good approximation of the PMF. We use the sum of dice as an example because it is easy to understand. Datasets in the real world often represent more exciting events.

Expectation

A random variable is fully represented by its probability mass function (PMF), which represents each of the values the random variable can take on, and the corresponding probabilities. A PMF can be a lot of information. Sometimes it is useful to summarize the random variable! The most common, and arguably the most useful, summary of a random variable is its "**Expectation**".

Definition: Expectation

The expectation of a random variable X , written $E[X]$ is the average of all the values the random variable can take on, each weighted by the probability that the random variable will take on that value.

$$E[X] = \sum_x x \cdot P(X = x)$$

Expectation goes by many other names: Mean, Weighted Average, Center of Mass, 1st Moment. All of which are calculated using the same formula.

Recall that $P(X = x)$, also written as $P(x)$, is the probability mass function of the random variable X . Here is code that calculates the expectation of the sum of two dice, based off the probability mass function:

```
def expectation_sum_two_dice():
    exp_sum_two_dice = 0
    # sum of dice can take on the values 2 through 12
    for x in range(2, 12 + 1):
        pr_x = pmf_sum_two_dice(x) # pmf gives Pr sum is x
        exp_sum_two_dice += x * pr_x
    return exp_sum_two_dice

def pmf_sum_two_dice(x):
    # Return the probability that two dice sum to x
    count = 0
    # Loop through all possible combinations of two dice
    for dice1 in range(1, 6 + 1):
        for dice2 in range(1, 6 + 1):
            if dice1 + dice2 == x:
                count += 1
    return count / 36 # There are 36 possible outcomes (6x6)
```

If we worked it out manually we would get that if X is the sum of two dice, $E[X] = 7$:

$$E[X] = \sum_x x \cdot P(X = x) = 2 \cdot \frac{1}{36} + 3 \cdot \frac{2}{36} + \cdots + 12 \cdot \frac{1}{36} = 7$$

7 is the "average" number you expect to get if you took the sum of two dice near infinite times. In this case it also happens to be the same as the mode, the most likely value of the sum of two dice, but this is not always the case!

Properties of Expectation

Property: Linearity of Expectation

$$E[aX + b] = aE[X] + b$$

Where a and b are constants and not random variables.

Property: Expectation of the Sum of Random Variables

$$E[X + Y] = E[X] + E[Y]$$

This is true regardless of the relationship between X and Y . They can be dependent, and they can have different distributions. This also applies with more than two random variables.

$$E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n E[X_i]$$

Property: Law of The Unconscious Statistician (LOTUS)

$$E[g(X)] = \sum_x g(x) P(X = x)$$

One can also calculate the expected value of a function $g(X)$ of a random variable X when one knows the probability distribution of X but one does not explicitly know the distribution of $g(X)$. This theorem has the humorous name of "the Law of the Unconscious Statistician" (LOTUS), because it is so useful that you should be able to employ it unconsciously.

Property: Expectation of a Constant

$$E[a] = a$$

Sometimes in proofs, you will end up with the expectation of a constant (rather than a random variable). For example what does the $E[5]$ mean? Since 5 is not a random variable, it does not change, and will always be 5, $E[5] = 5$.

Expectation of Sums Proof

One of the most useful properties of expectation is that the sum of expectation of random variables can be calculated by summing the expectations of each random variable on its own. Later in [Adding Random Variables](#) we will learn that computing the full probability mass function (or probability density function) when adding random variables is quite hard, especially when the random variables are dependent. However the expectation of sums is always decomposable:

$$E[X + Y] = E[X] + E[Y]$$

This very useful result is somewhat surprising. I believe that the best way to understand this result is via a proof. This proof, however, will use some concepts from the next section in the course reader, Probabilistic Models. Once you have learned how to think about random variables jointly you will be able to understand this proof. There is a section of the reader that goes over the proof in details and gives of visualization for why it is true! [Proof of Expectation of Sum of Random Variables](#).

Example of Law of Unconscious Statistician

The property of expectation:

$$E[g(X)] = \sum_x g(x) P(X = x)$$

is both useful and hard to understand just by reading the equation. It allows us to calculate the expectation of a function of any function applied to a random variable! One function that will turn out to be very useful when computing [Variance](#) is $E[X^2]$. According to the Law of Unconscious Statistician:

$$E[X^2] = \sum_x x^2 P(X = x)$$

In this case g is the squaring function. To calculate $E[X^2]$ we calculate expectation in a way similar to $E[X]$ with the exception that we square the value of x before multiplying by the probability mass function. Here is code that calculates $E[X^2]$ for the sum of two dice:

```
def expectation_sum_two_dice_squared():
    exp_sum_two_dice_squared = 0
    # sum of dice can take on the values 2 through 12
    for x in range(2, 12 + 1):
        pr_x = pmf_sum_two_dice(x) # pmf gives Pr(x)
        exp_sum_two_dice_squared += x**2 * pr_x
    return exp_sum_two_dice_squared
```

Expectation with Continuous Random Variables

Later in the course reader we will learn about continuous random variables. Expectation for those types of random variables is very similar. See [Continuos Random Variables](#) for details

Variance

Definition: Variance of a Random Variable

The variance is a measure of the "spread" of a random variable around the mean. Variance for a random variable, X , with expected value $E[X] = \mu$ is:

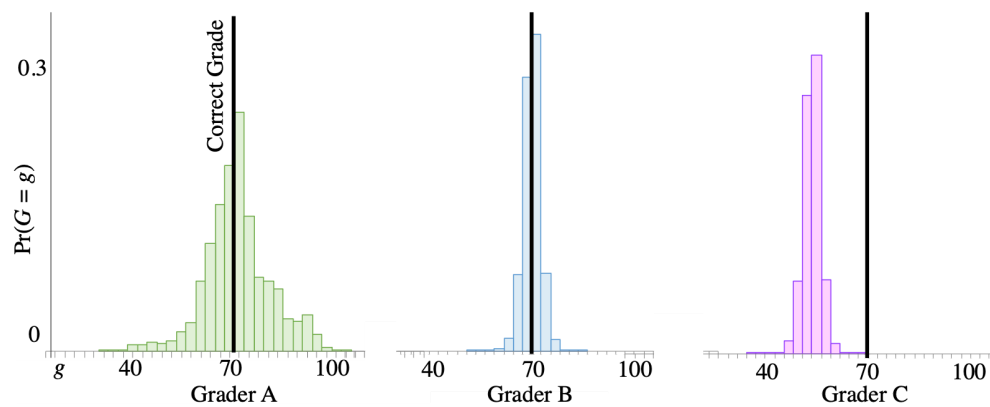
$$\text{Var}(X) = E[(X - \mu)^2]$$

Semantically, this is the average distance of a sample from the distribution to the mean. When computing the variance often we use a different (equivalent) form of the variance equation:

$$\text{Var}(X) = E[X^2] - E[X]^2$$

In the [last section](#) we showed that Expectation was a useful summary of a random variable (it calculates the "weighted average" of the random variable). One of the next most important properties of random variables to understand is variance: the measure of spread.

To start, let's consider probability mass functions for three sets of graders. When each of them grades an assignment, meant to receive a 70/100, they each have a probability distribution of grades that they could give.



Distributions of three types of peer graders. Data is from a massive online course.

The distribution for graders in group *C* has a different *expectation*. The average grade that they give when grading an assignment worth 70 is a 55/100. That is clearly not great! But what is the difference between graders *A* and *B*? Both of them have the same expected value (which is equal to the correct grade). The graders in group *A* have a higher "spread". When grading an assignment worth 70, they have a reasonable chance of giving it a 100, or of giving it a 40. Graders in group *B* have much less spread. Most of the probability mass is close to 70. You want graders like those in group *B*: in expectation they give the correct grade, and they have low spread. As an aside: scores in group *B* came from a probabilistic algorithm over peer grades.

Theorists wanted a number to describe spread. They invented variance to be the average of the distance between values that the random variable could take on and the mean of the random variable. There are many reasonable choices for the distance function, probability theorists chose squared deviation from the mean:

$$\text{Var}(X) = E[(X - \mu)^2]$$

Proof: $\text{Var}(X) = E[X^2] - E[X]^2$

It is much easier to compute variance using $E[X^2] - E[X]^2$. You certainly don't need to know why its an equivalent expression, but in case you were wondering, here is the proof.

$\text{Var}(X) = \text{E}[(X-\mu)^2]$	Note: $\mu = \text{E}[X]$
$= \sum_x (x - \mu)^2 P(x)$	Definition of Expectation
$= \sum_x (x^2 - 2\mu x + \mu^2) P(x)$	Expanding the square
$= \sum_x x^2 P(x) - 2\mu \sum_x x P(x) + \mu^2 \sum_x P(x)$	Propagating the sum
$= \sum_x x^2 P(x) - 2\mu \text{E}[X] + \mu^2 \sum_x P(x)$	Substitute def of expectation
$= \text{E}[X^2] - 2\mu \text{E}[X] + \mu^2 \sum_x P(x)$	LOTUS $g(x) = x^2$
$= \text{E}[X^2] - 2\mu \text{E}[X] + \mu^2$	Since $\sum_x P(x) = 1$
$= \text{E}[X^2] - 2 \text{E}[X]^2 + \text{E}[X]^2$	Since $\mu = \text{E}[X]$
$= \text{E}[X^2] - \text{E}[X]^2$	Cancellation

Standard Deviation

Variance is especially useful for comparing the "spread" of two distributions and it has the useful property that it is easy to calculate. In general a larger variance means that there is more deviation around the mean — more spread. However, if you look at the leading example, the units of variance are the square of points. This makes it hard to interpret the numerical value. What does it mean that the spread is 52 points²? A more interpretable measure of spread is the square root of Variance, which we call the Standard Deviation $\text{Std}(X) = \sqrt{\text{Var}(X)}$. The standard deviation of our grader is 7.2 points. In this example folks find it easier to think of spread in points rather than points². As an aside, the standard deviation is the average distance of a sample (from the distribution) to the mean, using the [euclidean distance](#) function.

Variance with Continuous Random Variables

Later in the course reader we will learn about continuous random variables. Variance for those types of random variables is very similar. See [Continuos Random Variables](#) for details.

Bernoulli Distribution

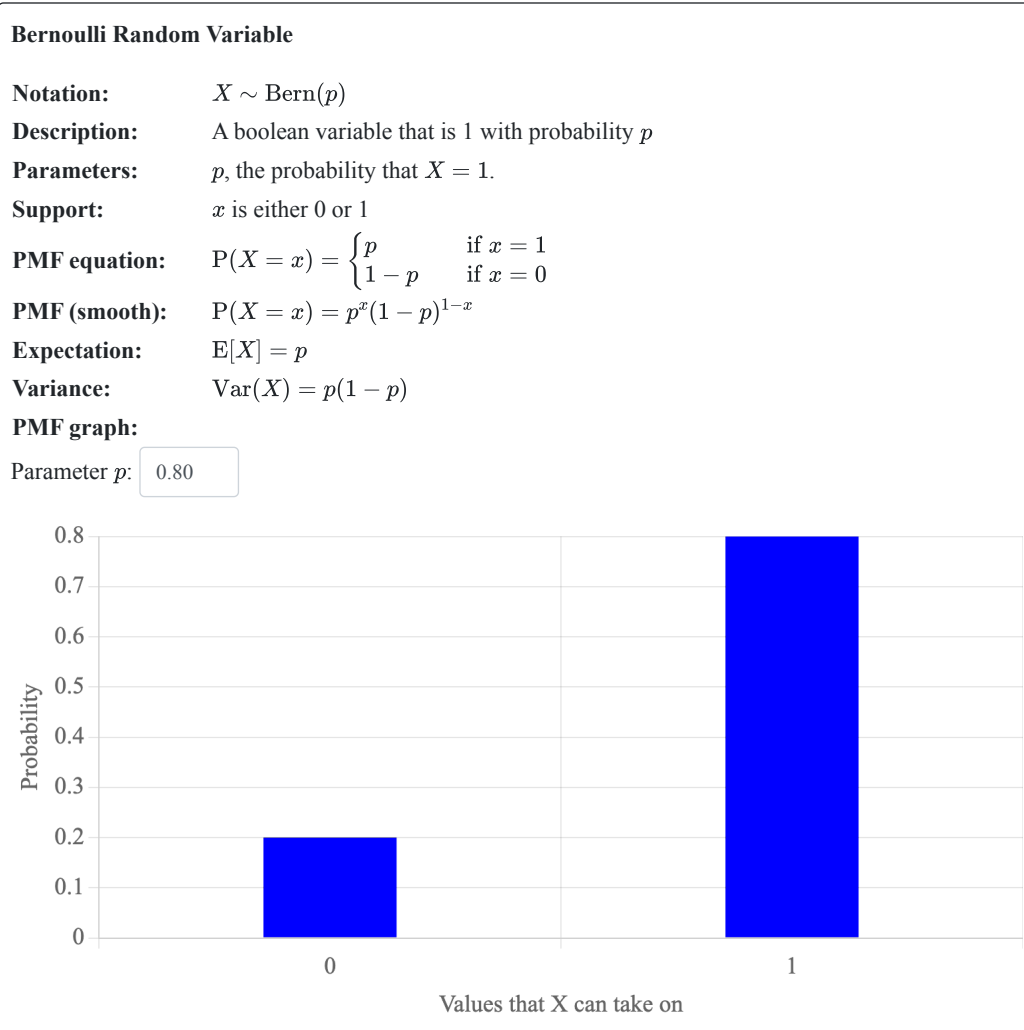
Parametric Random Variables

There are many classic and commonly-seen random variable abstractions that show up in the world of probability. At this point in the class, you will learn about several of the most significant parametric discrete distributions. When solving problems, if you can recognize that a random variable fits one of these formats, then you can use its pre-derived probability mass function (PMF), expectation, variance, and other properties. Random variables of this sort are called **parametric** random variables. If you can argue that a random variable falls under one of the studied parametric types, you simply need to provide parameters. A good analogy is a **class** in programming. Creating a parametric random variable is very similar to calling a constructor with input parameters.

Bernoulli Random Variables

A Bernoulli random variable (also called a *boolean* or *indicator* random variable) is the simplest kind of parametric random variable. It can take on two values, 1 and 0. It takes on a 1 if an experiment with probability p resulted in success and a 0 otherwise. Some example uses include a coin flip, a random binary digit, whether a disk drive crashed, and whether someone likes a Netflix movie. Here p is the parameter, but different instances of Bernoulli random variables might have different values of p .

Here is a full description of the key properties of a Bernoulli random variable. If X is declared to be a Bernoulli random variable with parameter p , denoted $X \sim \text{Bern}(p)$:



Because Bernoulli distributed random variables are parametric, as soon as you declare a random variable to be of type Bernoulli you automatically can know all of these pre-derived properties! Some of these properties are straightforward to prove for a Bernoulli. For example, you could have solved for expectation:

Proof: Expectation of a Bernoulli. If X is a Bernoulli with parameter p , $X \sim \text{Bern}(p)$:

$$\begin{aligned} E[X] &= \sum_x x \cdot P(X = x) && \text{Definition of expectation} \\ &= 1 \cdot p + 0 \cdot (1 - p) && X \text{ can take on values 0 and 1} \\ &= p && \text{Remove the 0 term} \end{aligned}$$

Proof: Variance of a Bernoulli. If X is a Bernoulli with parameter p , $X \sim \text{Bern}(p)$:

To compute variance, first compute $E[X^2]$:

$$\begin{aligned} E[X^2] &= \sum_x x^2 \cdot P(X = x) && \text{LOTUS} \\ &= 0^2 \cdot (1 - p) + 1^2 \cdot p \\ &= p \\ \text{Var}(X) &= E[X^2] - E[X]^2 && \text{Def of variance} \\ &= p - p^2 && \text{Substitute } E[X^2] = p, E[X] = p \\ &= p(1 - p) && \text{Factor out } p \end{aligned}$$

Indicator Random Variable

Definition: Indicator Random Variable

An indicator variable is a Bernoulli random variable which takes on the value 1 if an underlying event occurs, and 0 otherwise.

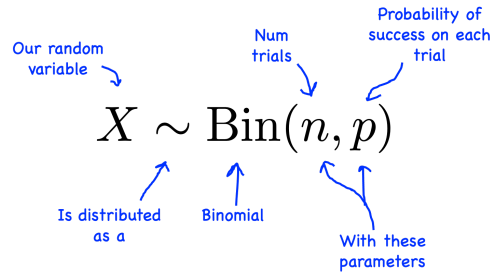
Indicator random variables are a convenient way to convert the "true/false" outcome of an event into a number. That number may be easier to incorporate into an equation. See the [binomial expectation derivation](#) for an example.

A random variable I is an indicator variable for an event A if $I = 1$ when A occurs and $I = 0$ if A does not occur. Indicator random variables are Bernoulli random variables, with $p = P(A)$. I is a common choice of name for an indicator random variable.

Here are some properties of indicator random variables: $P(I = 1) = P(A)$ and $E[I] = P(A)$.

Binomial Distribution

In this section, we will discuss the binomial distribution. To start, imagine the following example. Consider n independent trials of an experiment where each trial is a "success" with probability p . Let X be the number of successes in n trials. This situation is truly common in the natural world, and as such, there has been a lot of research into such phenomena. Random variables like X are called binomial random variables. If you can identify that a process fits this description, you can inherit many already proved properties such as the PMF formula, expectation, and variance!



Here are a few examples of binomial random variables:

- # of heads in n coin flips
- # of 1's in randomly generated length n bit string
- # of disk drives crashed in 1000 computer cluster, assuming disks crash independently

Binomial Random Variable

Notation: $X \sim \text{Bin}(n, p)$

Description: Number of "successes" in n identical, independent experiments each with probability of success p .

Parameters: $n \in \{0, 1, \dots\}$, the number of experiments.
 $p \in [0, 1]$, the probability that a single experiment gives a "success".

Support: $x \in \{0, 1, \dots, n\}$

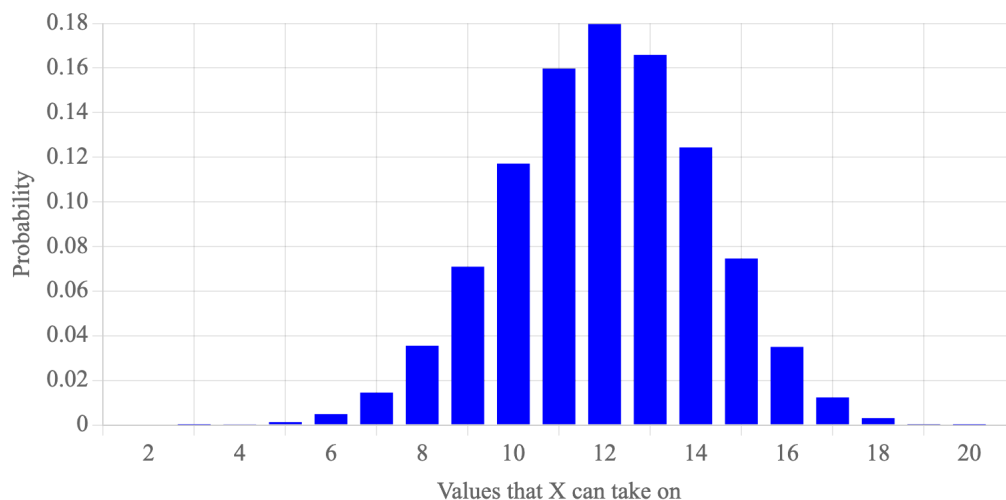
PMF equation: $P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$

Expectation: $E[X] = n \cdot p$

Variance: $\text{Var}(X) = n \cdot p \cdot (1 - p)$

PMF graph:

Parameter n : Parameter p :



One way to think of the binomial is as the sum of n [Bernoulli](#) variables. Say that $Y_i \sim \text{Bern}(p)$ is an indicator Bernoulli random variable which is 1 if experiment i is a success. Then if X is the total number of successes in n experiments, $X \sim \text{Bin}(n, p)$:

$$X = \sum_{i=1}^n Y_i$$

Recall that the outcome of Y_i will be 1 or 0, so one way to think of X is as the sum of those 1s and 0s.

Binomial PMF

The most important property to know about a binomial is its [PMF function](#):

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Probability Mass Function
for a Binomial

↑
Probability that our
variable takes on the
value k

Recall, we derived this formula in Part 1. There is a complete example on the probability of k heads in n coin flips, where each flip is heads with probability 0.5: [Many Coin Flips](#). To briefly review, if you think of each experiment as being distinct, then there are $\binom{n}{k}$ ways of permuting k successes from n experiments. For any of the mutually exclusive permutations, the probability of that permutation is $p^k \cdot (1 - p)^{n-k}$.

The name binomial comes from the term $\binom{n}{k}$ which is formally called the binomial coefficient.

Expectation of Binomial

There is an easy way to calculate the expectation of a binomial and a hard way. The easy way is to leverage the fact that a binomial is the sum of Bernoulli [indicator](#) random variables. $X = \sum_{i=1}^n Y_i$ where Y_1 is an indicator of whether the i th experiment was a success: $Y_i \sim \text{Bern}(p)$. Since the [expectation of the sum](#) of random variables is the sum of expectations, we can add the expectation, $E[Y_i] = p$, of each of the Bernoulli's:

$$\begin{aligned}
 E[X] &= E\left[\sum_{i=1}^n Y_i\right] && \text{Since } X = \sum_{i=1}^n Y_i \\
 &= \sum_{i=1}^n E[Y_i] && \text{Expectation of sum} \\
 &= \sum_{i=1}^n p && \text{Expectation of Bernoulli} \\
 &= n \cdot p && \text{Sum } n \text{ times}
 \end{aligned}$$

The hard way is to use the definition of expectation:

$$\begin{aligned}
 E[X] &= \sum_{i=0}^n i \cdot P(X = i) && \text{Def of expectation} \\
 &= \sum_{i=0}^n i \cdot \binom{n}{i} p^i (1 - p)^{n-i} && \text{Sub in PMF} \\
 &\dots && \text{Many steps later} \\
 &= n \cdot p
 \end{aligned}$$

Binomial Distribution in Python

As you might expect, you can use binomial distributions in code. The standardized library for binomials is `scipy.stats.binom`.

One of the most helpful methods that this package provides is a way to calculate the PMF. For example, say $X \sim \text{Bin}(n = 5, p = 0.6)$ and you want to find $P(X = 2)$ you could use the following code:

```
from scipy import stats

# define variables for x, n, and p
n = 5
p = 0.6
x = 2

# use scipy to compute the pmf
p_x = stats.binom.pmf(x, n, p)

# use the probability for future work
print(f'P(X = {x}) = {p_x}')
```

Console:

```
P(X = 2) = 0.2304
```

Another particularly helpful function is the ability to generate a random sample from a binomial. For example, say $X \sim \text{Bin}(n = 10, p = 0.3)$ represents the number of requests to a website. We can draw 100 samples from this distribution using the following code:

```
from scipy import stats

# define variables for x, n, and p
n = 10
p = 0.3
x = 2

# use scipy to compute the pmf
samples = stats.binom.rvs(n, p, size=100)

# use the probability for future work
print(samples)
```

Console:

```
[4 5 3 1 4 5 3 1 4 6 5 6 1 2 1 1 2 3 2 5 2 2 2 4 4 2 2 3 6 3 1 1 4 2 6 2 4
 2 3 3 4 2 4 2 4 5 0 1 4 3 4 3 3 1 3 1 1 2 2 2 3 5 3 3 3 2 1 3 2 1 2 3 3
 4 5 1 3 7 1 4 1 3 3 4 4 1 2 4 4 0 2 4 3 2 3 3 1 1 4]
```

You might be wondering what a random sample is! A random sample is a randomly chosen assignment for our random variable. Above we have 100 such assignments. The probability that value x is chosen is given by the PMF: $P(X = x)$. You will notice that even though 8 is a possible assignment to the binomial above, in 100 samples we never saw the value 8. Why? Because $P(X = 8) \approx 0.0014$. You would need to draw 1,000 samples before you would expect to see an 8.

There are also functions for getting the mean, the variance, and more. You can read the [scipy.stats.binom documentation](#), especially the list of methods.